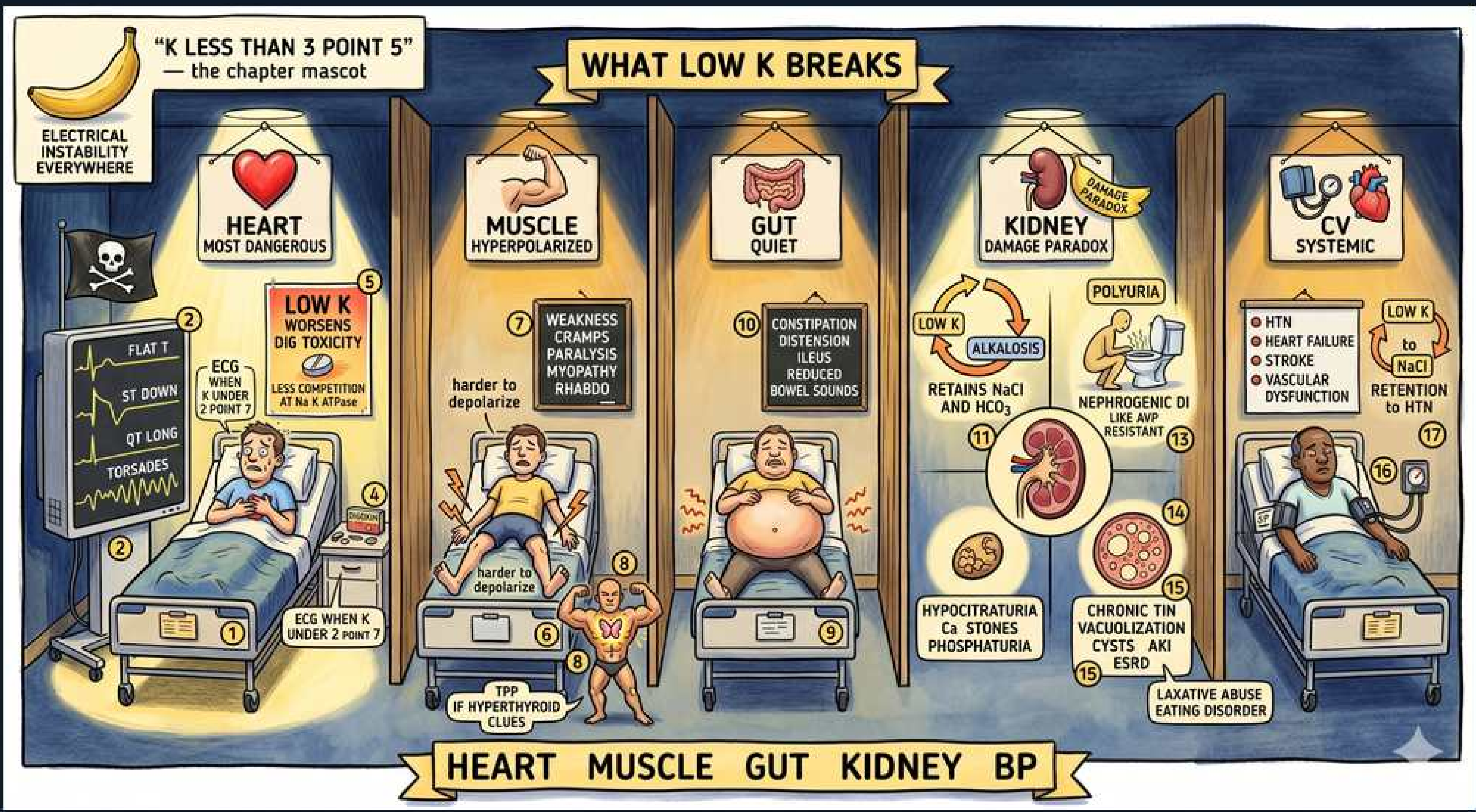
# Hypokalemia — clinical features



## 1. Low K breaks 5 systems

**WHAT YOU SEE**

Banner: WHAT LOW K BREAKS — heart, muscle, gut, kidney, BP

**WHAT IT ENCODES**

Hypokalemia (serum K <3.5 mEq/L; severe <2.5) is a state of membrane electrical instability hitting five systems: heart (arrhythmia), skeletal muscle (weakness/rhabdo), GI smooth muscle (ileus), kidney (nephrogenic DI), and cardiovascular/BP (HTN). Most patients are asymptomatic until K <3.0; symptoms track severity and rate of fall. Mnemonic: 'Heart, Muscle, Gut, Kidney, BP.'

**BOARD TRAP**

WRONG: assuming a mildly low K is harmless in everyone. Discriminator — even modest hypokalemia is dangerous in patients on digoxin, with cardiac disease, or with QT-prolonging drugs; risk is context-dependent, not just number-dependent.

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## 2. Heart — most dangerous

**WHAT YOU SEE**

Heart door, skull, MOST DANGEROUS

**WHAT IT ENCODES**

Cardiac toxicity is the lethal effect and the reason hypokalemia is an emergency. Low K delays repolarization and increases automaticity → atrial and ventricular ectopy, AF, VT/VF, and Torsades, especially with concomitant digoxin, hypomagnesemia, or QT-prolonging drugs. Continuous ECG monitoring is mandatory in severe or symptomatic cases.

**BOARD TRAP**

WRONG: waiting for muscle weakness before worrying. Discriminator — fatal arrhythmia can occur before any neuromuscular symptom; cardiac monitoring and prompt K repletion take priority in severe/symptomatic hypokalemia.

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## 3. ECG changes

**WHAT YOU SEE**

ECG monitor: flat T, ST down, QT long, U wave, torsades; ECG when K under 2.7

**WHAT IT ENCODES**

ECG sequence with falling K: flattened/inverted T waves, ST-segment depression, prominent U waves, apparent QT (really QU) prolongation, then VT/Torsades. Get an ECG when K <2.7. The U wave (a deflection after the T) is the signature finding. Coexisting hypomagnesemia amplifies all of these.

**BOARD TRAP**

WRONG ANSWER: calling 'peaked T waves' a sign of hypokalemia. Why wrong — peaked/tented T waves are HYPERkalemia. Discriminator — hypokalemia = flat T + U wave + ST depression; hyperkalemia = peaked T + wide QRS. Don't swap them.

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## 4. Low K + digoxin toxicity

**WHAT YOU SEE**

LOW K TOXICITY box: low K worsens digoxin toxicity, K competes at site

**WHAT IT ENCODES**

Potassium and digoxin compete for the same binding site on the Na/K-ATPase. Low K means less competition → more digoxin bound → enhanced effect and toxicity even at therapeutic digoxin levels. So hypokalemia (and hypomagnesemia) precipitates digoxin toxicity arrhythmias; correcting K is part of treating dig toxicity.

**BOARD TRAP**

WRONG: in a digoxin patient with new arrhythmia, focusing only on the digoxin level. Discriminator — a 'therapeutic' dig level can still be toxic if K is low; check and correct K (and Mg) — hypokalemia is the hidden trigger.

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## 5. Muscle — hyperpolarized

**WHAT YOU SEE**

Muscle door: hyperpolarized

**WHAT IT ENCODES**

Low extracellular K raises the K gradient and hyperpolarizes the resting membrane (makes it more negative), so a larger stimulus is needed to reach threshold → impaired depolarization and muscle weakness. This is the unifying mechanism behind the neuromuscular and smooth-muscle features.

**BOARD TRAP**

WRONG: assuming low K, like low Ca, causes hyperexcitability/tetany. Discriminator — hypokalemia HYPERpolarizes → weakness/hypotonia; it is hypocalcemia that causes tetany. Different ion, opposite clinical picture.

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## 6. Weakness / paralysis / rhabdo

**WHAT YOU SEE**

Weakness, cramps, paralysis, myopathy, rhabdo; harder to depolarize

**WHAT IT ENCODES**

Skeletal-muscle effects scale with severity: cramps and fatigue → proximal then ascending weakness → flaccid paralysis (can involve respiratory muscles) at K <2.5. Severe hypokalemia impairs exercise-induced vasodilation in muscle → ischemia → rhabdomyolysis, which then releases K and can mask the deficit.

**BOARD TRAP**

WRONG: dismissing severe hypokalemia as 'just mild weakness.' Discriminator — K <2.5 can cause respiratory muscle paralysis and rhabdomyolysis; this is a medical emergency requiring IV K and monitoring, not oral repletion alone.

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## 7. Periodic paralysis clue

**WHAT YOU SEE**

Small figure: TPP if hyperthyroid clues

**WHAT IT ENCODES**

Episodic flaccid paralysis with hypokalemia from transcellular SHIFT (not total-body loss) = periodic paralysis: hypokalemic familial (channelopathy) or thyrotoxic periodic paralysis (TPP, classically young Asian males after carbohydrate/exercise). Treat cautiously with low-dose K (and beta-blocker in TPP) because K rebounds as it shifts back out.

**BOARD TRAP**

WRONG: aggressively repleting K in periodic paralysis as if it were a true deficit. Discriminator — the K is shifted, not lost; over-repletion causes rebound HYPERkalemia when cells release it. Give small doses and treat the cause (e.g. control hyperthyroidism).

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### 8. Gut — quiet/ileus

**WHAT YOU SEE**

Gut door: quiet

**WHAT IT ENCODES**

GI smooth muscle also hyperpolarizes → hypomotility: constipation, abdominal distension, decreased/absent bowel sounds, and paralytic ileus. The gut goes 'quiet' — the smooth-muscle parallel to skeletal weakness.

**BOARD TRAP**

WRONG: attributing a stubborn postoperative ileus solely to opioids/surgery. Discriminator — check the K in any unexplained or refractory ileus; hypokalemia is a correctable cause of gut paralysis.

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### 9. GI symptoms

**WHAT YOU SEE**

Constipation, distension, ileus, reduced bowel sounds

**WHAT IT ENCODES**

The constellation — constipation, distension, ileus, reduced bowel sounds — all stems from smooth-muscle hypomotility. Severe ileus can in turn impair oral intake and worsen the electrolyte picture, so repletion plus addressing the cause is needed.

**BOARD TRAP**

WRONG: continuing to push oral potassium in a patient with an ileus. Discriminator — with ileus/poor gut motility, enteral absorption is unreliable; use IV repletion until the gut recovers.

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### 10. Kidney — damage paradox

**WHAT YOU SEE**

Kidney door: damage paradox, polyuria

**WHAT IT ENCODES**

Renal effects: hypokalemia causes ADH resistance (nephrogenic DI → polyuria/polydipsia) and increased renal ammoniagenesis plus enhanced bicarbonate reabsorption, which generates/maintains a metabolic alkalosis. Chronically it injures the tubules. The 'paradox': low K both results from and then perpetuates a renal-driven alkalosis.

**BOARD TRAP**

WRONG: ignoring the K when working up new polyuria. Discriminator — hypokalemia is a reversible cause of nephrogenic DI; it should be on the differential for polyuria alongside hypercalcemia and lithium.

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### 11. Nephrogenic DI / polyuria

**WHAT YOU SEE**

Nephrogenic DI ADH resistant, retains HCO3, polyuria

**WHAT IT ENCODES**

Chronic hypokalemia downregulates aquaporin-2 channels → kidney cannot concentrate urine despite ADH (ADH-resistant nephrogenic DI) → polyuria, nocturia, polydipsia. Simultaneously increased ammoniagenesis and HCO3 retention help sustain a metabolic alkalosis, linking the renal and acid-base effects.

**BOARD TRAP**

WRONG: giving desmopressin for the polyuria of hypokalemic nephrogenic DI. Discriminator — nephrogenic DI is ADH-RESISTANT, so DDAVP won't work; the fix is correcting the potassium (and removing causes).

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### 12. Chronic TIN / cysts / ESRD

**WHAT YOU SEE**

Chronic tubulointerstitial nephritis, vacuolization, cysts, AKI/ESRD

**WHAT IT ENCODES**

Prolonged severe hypokalemia (classically chronic laxative abuse, eating disorders, or long-standing diuretic use) causes tubular vacuolization, chronic tubulointerstitial nephritis, renal cysts, and progressive CKD that can reach ESRD. This is the structural end-stage of the renal damage paradox.

**BOARD TRAP**

WRONG: assuming hypokalemic kidney injury is always fully reversible. Discriminator — brief hypokalemia is reversible, but chronic hypokalemic nephropathy (laxative/eating-disorder patients) can produce permanent tubulointerstitial damage and CKD.

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### 13. CV — systemic / HTN

**WHAT YOU SEE**

CV door: HTN, heart failure, stroke, vascular dysfunction; low K to HTN

**WHAT IT ENCODES**

Low potassium promotes renal sodium retention and impairs vasodilation → higher blood pressure; population data link low-K/high-Na diets to HTN, heart failure, and stroke. Conversely, increasing dietary K lowers BP (DASH diet rationale). In mineralocorticoid-excess states, HTN and hypokalemia appear together.

**BOARD TRAP**

WRONG ANSWER: workup of HTN + spontaneous hypokalemia stopping at 'essential hypertension.' Discriminator — unprovoked hypokalemia with HTN points to mineralocorticoid excess (primary hyperaldosteronism, etc.); pursue renin/aldosterone testing rather than calling it essential.

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